

Mastaan Uppal

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🏠 Surrey, B.C.

🌐 www.linkedin.com/in/mastaan-uppal

🔗 <https://mastaanuppal-dotcom.github.io/Portfolio/>

Technical Skills

Software

VHDL, Intel FPGA, Digital Logic
Python, C++, Java, MATLAB, JavaScript.
PCB design using EAGLE.
LTSpice for schematic design and testing.
Microsoft Office for organizing and analyzing data.
AutoCAD

Hardware

PCB assembly and soldering (through-hole and surface mount).
Microcontroller interfacing with Arduino.
Reading and interpreting circuit schematics.
SolidWorks for 3D modeling.
Use of oscilloscopes, multimeters, and DC power supplies in lab settings.

Technical Projects

Custom Motorized Coffee Grinder: Engineering Design and Fabrication

Sep 2024 – Dec 2024

- Designed and built a motorized grinder using a 24V DC motor and precision burrs
- Constructed a custom wood base and mechanical couplings for stability and alignment
- Prototyped a grind-by-weight system for consistent output
- Optimized grind quality through iterative testing of motor load and durability

PCB Assembly and Soldering: Fundamentals of Electronics

Nov 2024 – Dec 2024

- Assembled and soldered components onto a prefabricated PCB using precision techniques
- Interpreted schematics to place components accurately and minimize damage
- Verified board functionality via systematic testing and troubleshooting
- Gained hands-on experience with PCB layouts and electronics lab tools

Work Experience

Field Engineer - Co-op

Sep 2025 – April 2026

Flatiron Dragados, Richmond, B.C

- Contributed to the **16 km Surrey–Langley SkyTrain expansion** by coordinating electrical and roadway IFC design, producing detailed markups in Bluebeam and supporting high-quality RFI submissions to resolve field and design discrepancies
- Leveraged **Propeller Aero and CAD tools** to process survey data, analyze site conditions, and optimize layouts for curbs, gutters, and roadway features, improving constructability for field crews
- Designed and implemented **Excel-based tracking systems** with automated formulas to streamline quantity tracking, scheduling, and field reporting workflows
- Managed coordination across **5 subcontractors (>\$20M contracts each)**, supporting progress tracking, communication, and alignment with project deliverables

Engineering Assistant – Marine Design

Jun 2024 - Sep 2024

Autoship Systems Corporation, Vancouver, B.C

- Used **AutoCAD** to trace and modify ship design schematics for naval architecture projects
- Supported engineers by preparing accurate design layers and technical drawings
- Gained experience interpreting complex marine blueprints and design standards

Education

Simon Fraser University BASC: Electronics Engineering

Sep 2024 – Present